Exercises learning task 13

2026 – 2027

Exercise 8.4: A confidence interval estimate is desired for the gain in a circuit on a semiconductor device. Assume that gain is normally distributed with standard deviation $\sigma = 25$.

(a) Find a $95\,\%$ CI for μ when $n = 10$ and $\bar{x} = 1000$.

(b) Find a $95\,\%$ CI for μ when $n = 25$ and $\bar{x} = 1000$.

(c) Find a $99\,\%$ CI for μ when $n = 10$ and $\bar{x} = 1000$.

(d) Find a $99\,\%$ CI for μ when $n = 25$ and $\bar{x} = 1000$.

(e) How does the length of the CIs computed above change with the changes in sample size and confidence level?

Exercise 8.7: Consider the gain estimation problem in Exercise 8-4.

(a) How large must n be if the length of the $95\,\%$ CI is to be 30?

(b) How large must n be if the length of the $99\,\%$ CI is to be 30?

Exercise 8.9: Suppose that $n = 100$ random samples of water from a freshwater lake were taken and the calcium concentration ($\mathrm{mg\,L^{-1}}$) measured. A $95\,\%$ CI on the mean calcium concentration is $0.52 \leq \mu \leq 0.74$.

(a) Would a $99\,\%$ CI calculated from the same sample data be longer or shorter?

(b) Consider the following statement: "There is a $95\,\%$ chance that μ is between 0.52 and 0.74." Is this statement correct? Explain your answer.

(c) Consider the following statement: "If $n = 100$ random samples of water from the lake were taken and the $95\,\%$ CI on μ computed, and this process were repeated 1000 times, 950 of the CIs would contain the true value of μ." Is this statement correct? Explain your answer.

Exercise 8.11: The yield of a chemical process is being studied. From previous experience, yield is known to be normally distributed and $\sigma = 3$. The past five days of plant operation have resulted in the following percent yields: 91.6, 87.34, 90.8, 89.65, and 91.3. Find a $95\,\%$ two-sided confidence interval on the true mean yield.

Exercise 8.17: The life in hours of a 75-watt light bulb is known to be normally distributed with $\sigma = 20$ hours. Suppose you wanted the total width of the two-sided confidence interval on mean life to be six hours at $95\,\%$ confidence. What sample size should be used?

Exercise 8.29: A postmix beverage machine is adjusted to release a certain amount of syrup into a chamber where it is mixed with carbonated water. A random sample of 25 beverages was found to have a mean syrup content of $\bar{x} = 30\,\mathrm{mL}$. Suppose the standard deviation is known and equal to $\sigma = 0.5\,\mathrm{mL}$. Find a $95\,\%$ CI on the mean volume of syrup dispensed.

Exercise 7.63: Let $f(x) = (1/\theta)x^{(1-\theta)/\theta}$, $0 < x < 1$, and $0 < \theta < \infty$. Show that $\hat{\Theta} = -(1/n)\sum_{i=1}^{n} \ln(X_i)$ is the maximum likelihood estimator for θ and that $\hat{\Theta}$ is an unbiased estimator for θ.